# Ethanol Blending in Gasoline – El Salvador

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U.S. GRAINS & BIOPRODUCTS COUNCIL

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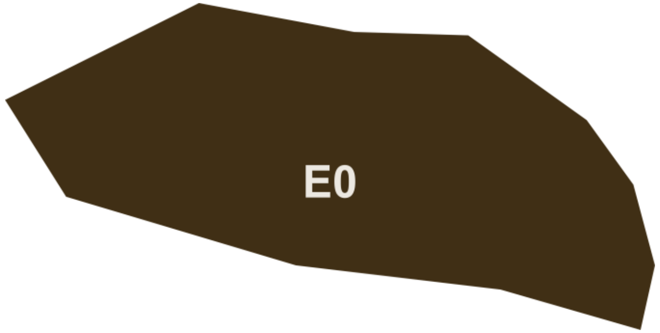
# Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.

# Ethanol Blending in Gasoline - El Salvador

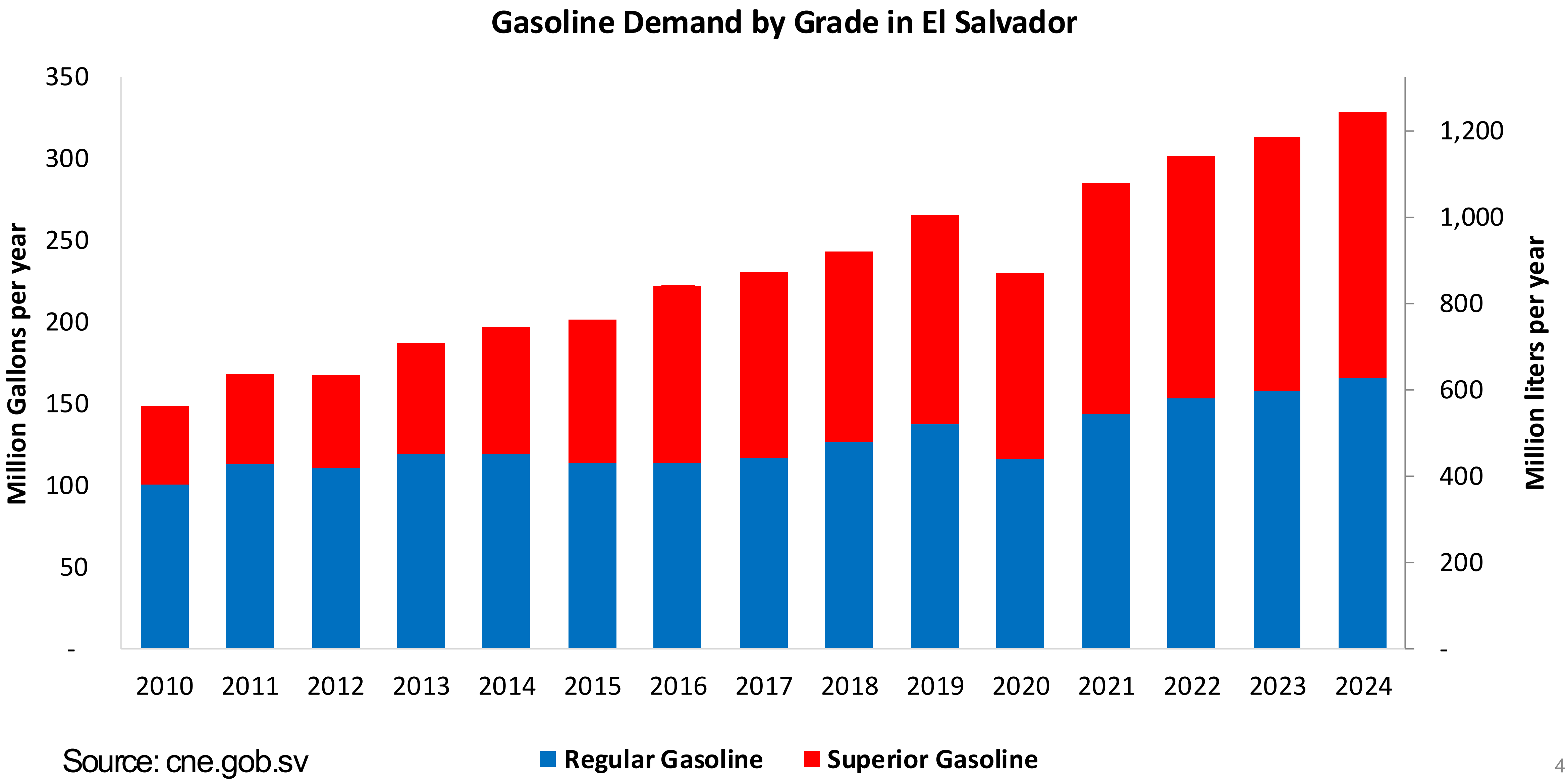



In 2024, gasoline demand was 328 million gallons (1,243 million liters). Regular gasoline RON 91 represented 51% of the volume consumed while Superior gasoline RON 95 reached 49%. Gasoline is supplied only by imports, distributed mainly by Chevron, Puma and Uno companies and imported mainly from the United States.

El Salvador has no national mandate for ethanol. It has the potential to produce ethanol from maize and sugar cane. In the last year there has been an interest for its use.

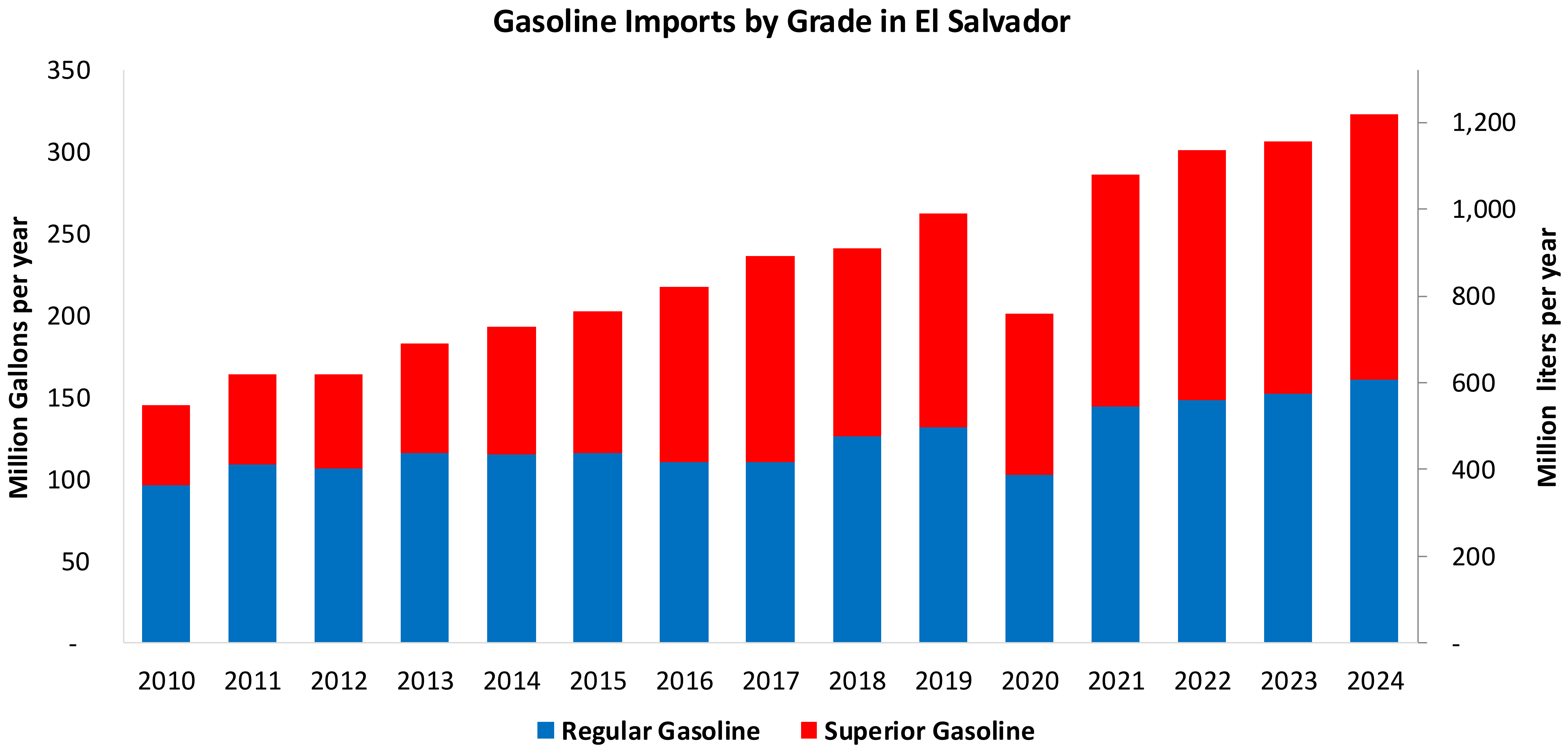
Sources: cne.gob.sv

# Gasoline Demand in El Salvador




Source: cne.gob.sv

# Gasoline Imports in El Salvador




Source: cne.gob.sv

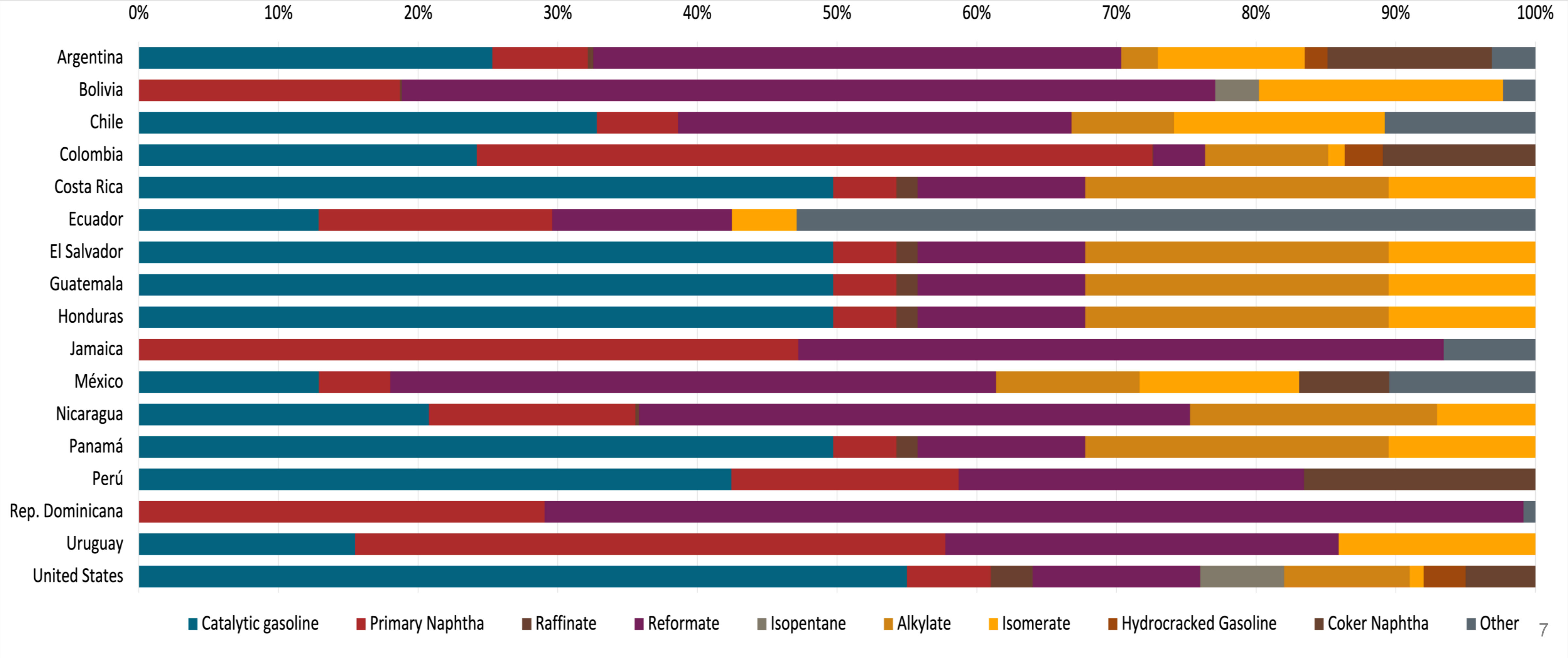
# Gasoline Quality in El Salvador

| Name | RTCA 75.01.19:19 | RTCA 75.01.20:19 | EN 228:2012 + A1:2017 (Euro 6 enabling) | | | |
|---|---|---|---|---|---|---|
| Implementation Date | 2021 | 2021 | 2017 | | | |
| Applicability | Whole country | Whole country | All countries | | | |
| Selected Grade | Gasoline Regular | Gasoline Premium | RON 95 E5 | RON 95 E10 | RON 98 E5 | RON 98 E10 |
| Benzene Content | 5% v/v | 5% v/v | < 1 %v/v | < 1 %v/v | < 1 %v/v | < 1 %v/v |
| Aromatics | 50% v/v | 50% v/v | < 35 %v/v | < 35 %v/v | < 35 %v/v | < 35 %v/v |
| Olefins | 30% v/v | 30% v/v | < 18 %v/v | < 18 %v/v | < 18 %v/v | < 18 %v/v |
| Lead Content | < 0,013 g/l | < 0,013 g/l | < 5 mg/l | < 5 mg/l | < 5 mg/l | < 5 mg/l |
| Manganese | 2,0% v/v | 2,0% v/v | < 2,0 mg/l | < 2,0 mg/l | < 2,0 mg/l | < 2,0 mg/l |
| RON | > 91 | > 95 | > 95 | > 95 | > 98 | > 98 |
| MON | - | - | > 85 | > 88 | > 85 | > 88 |
| AKI | | | | | | |
| Sulfur Content | < 500 mg/kg | < 500 mg/kg | < 10 mg/kg | < 10 mg/kg | < 10 mg/kg | < 10 mg/kg |
| Oxygen Content | 0,7% v/v | 0,7% v/v | <2,7 % m/m | <3,7 % m/m | <2,7 % m/m | <3,7 % m/m |
| Ethanol (EtOH) | - | - | <5 %v/v | <10 %v/v | <5 %v/v | <10 %v/v |
| RVP 37.8°C (Summer) | < 69 kPa | < 69 kPa | <> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive | | | |
| RVP 37.8 °C(Winter) | | | | | | |
| RVP 37.8°C (Transition) | | | | | | |
| MTBE | - | - | - | - | - | - |
| Ethers 5 or more C Atoms | - | - | Based on oxygen content | <22 %v/v | Based on oxygen content | <22 %v/v |

Source: RTCA

# Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



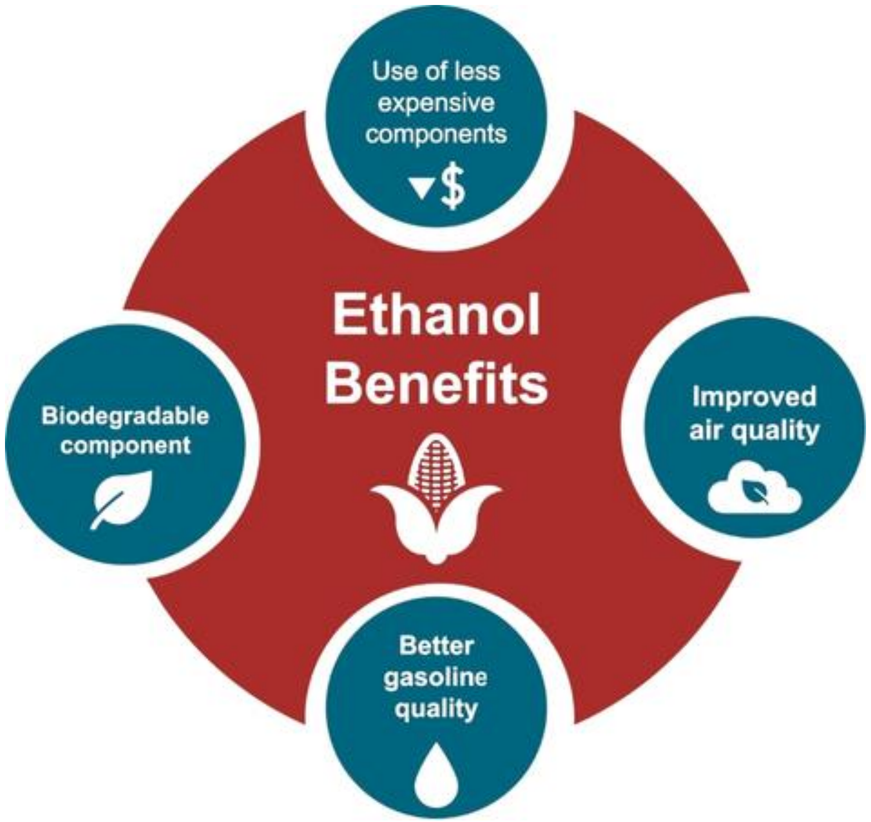
# Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

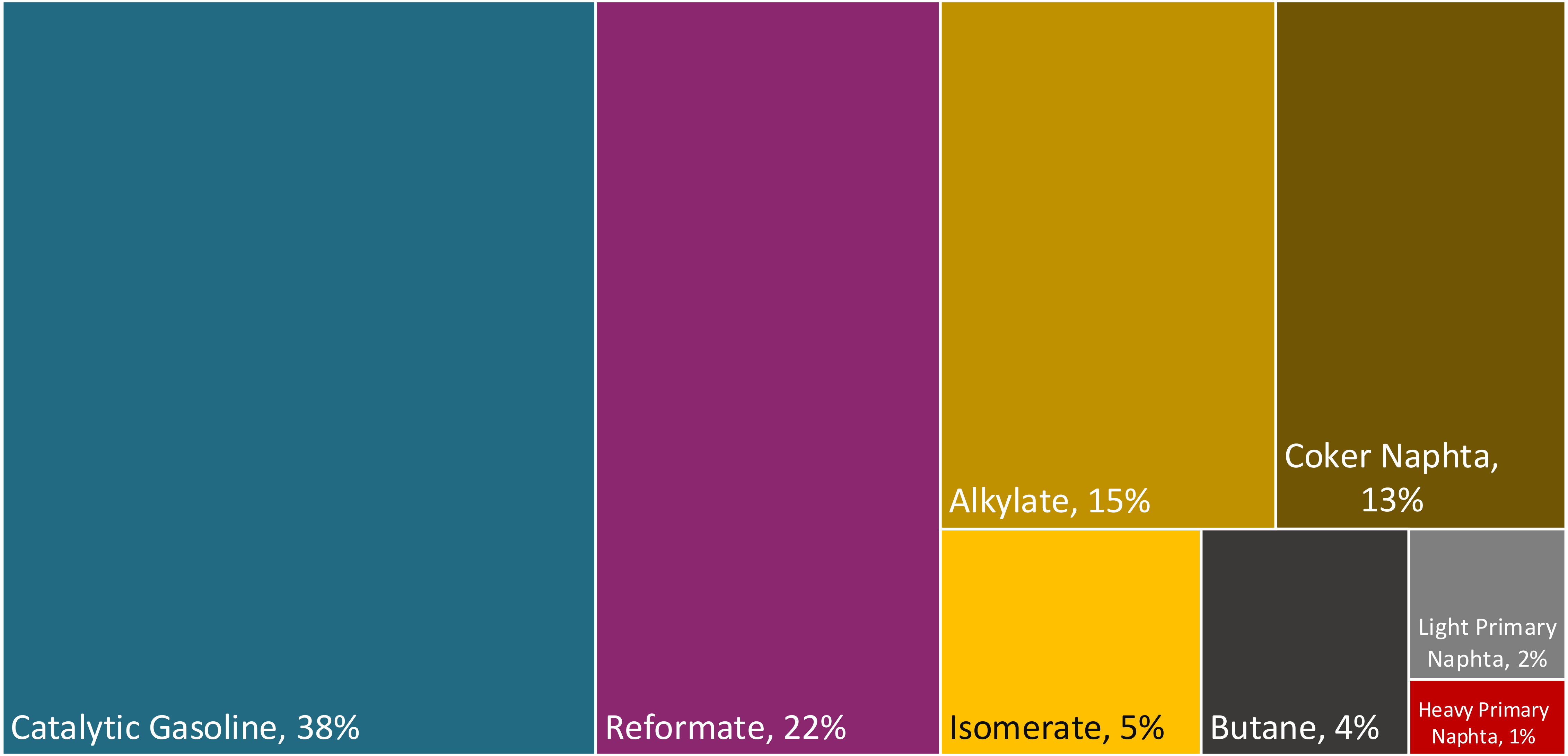
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

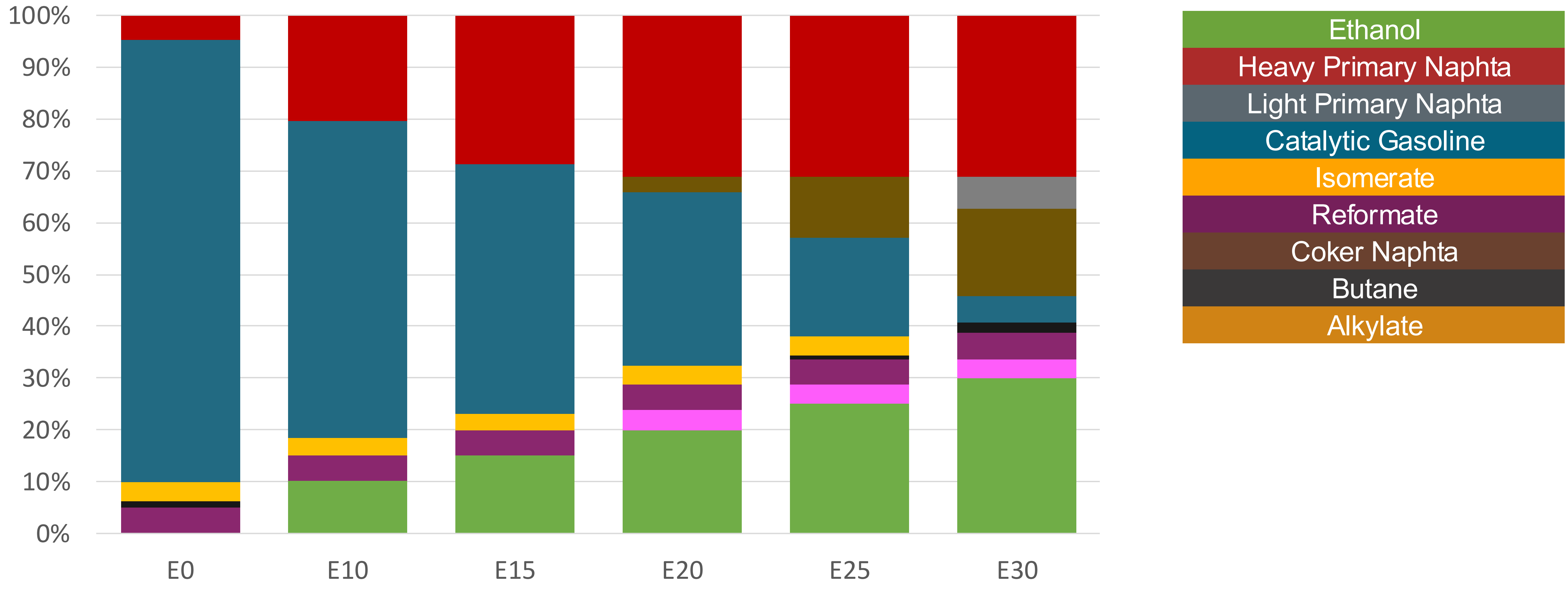


# Blending Components- El Salvador



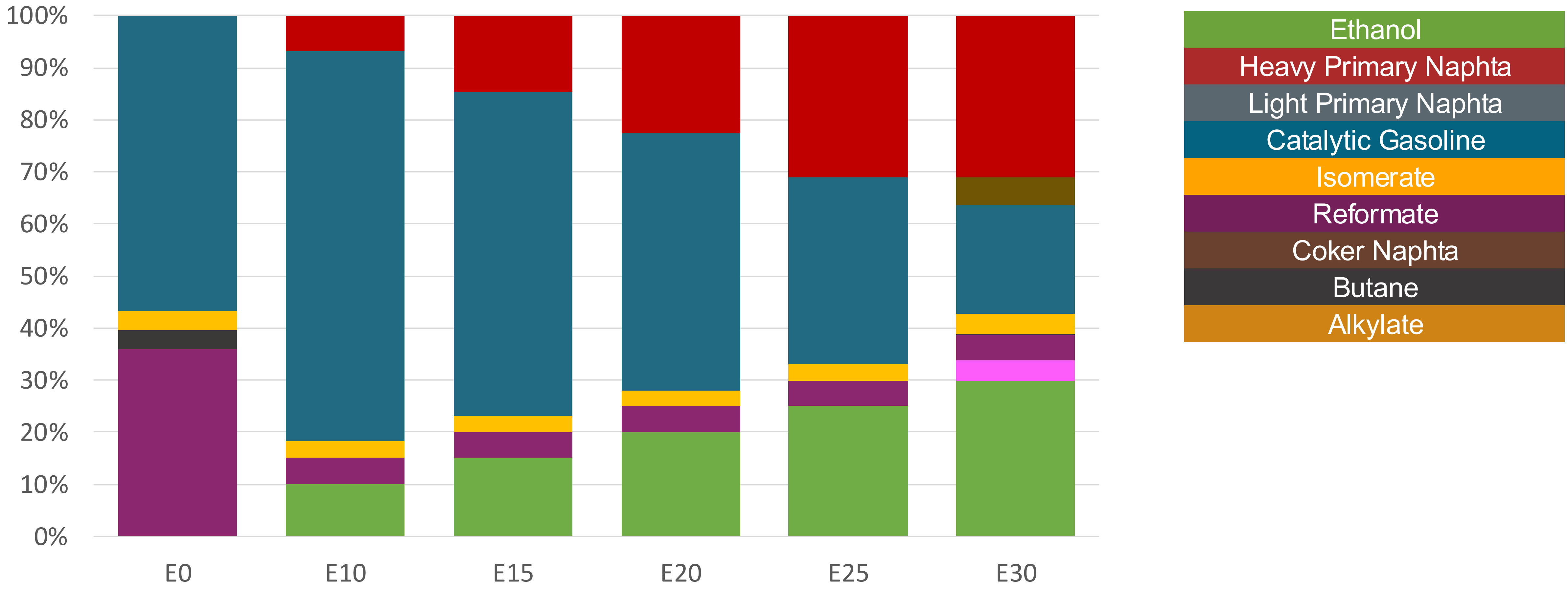

Source: Faro90

# El Salvador – Regular – Constant Octane



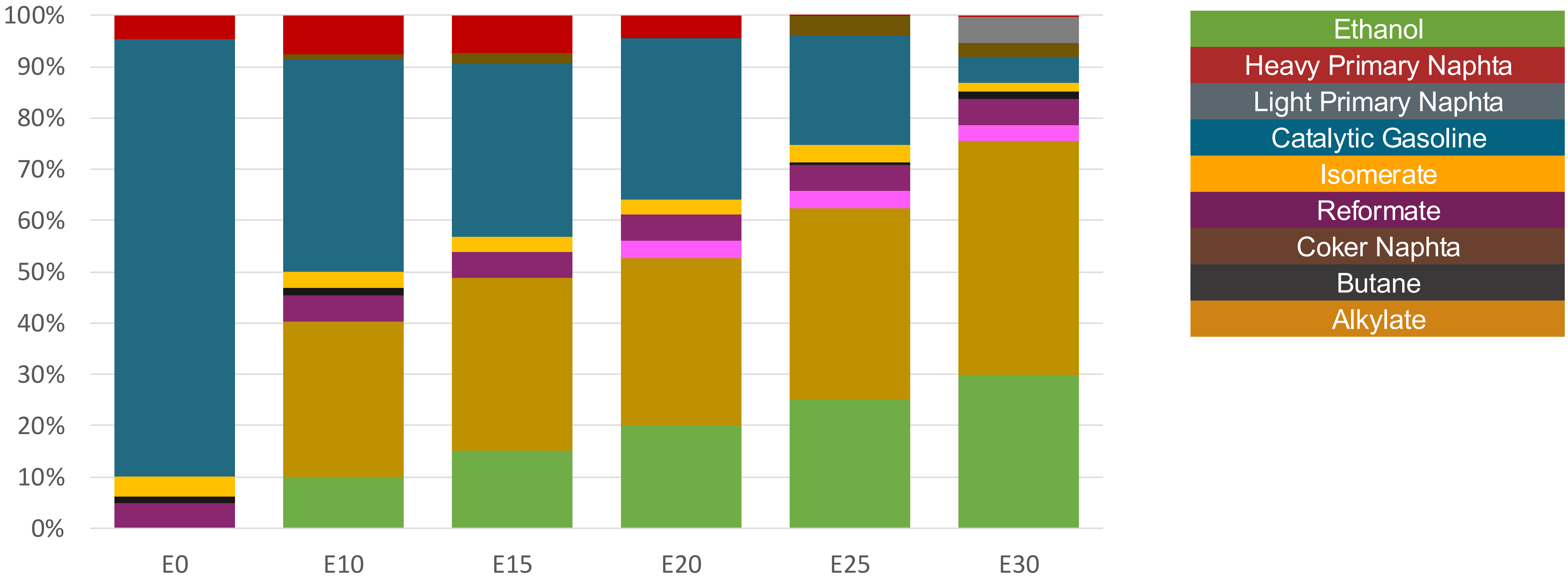

| | | | | | | |
|---|---|---|---|---|---|---|
| **Octane (RON)** | 91.0 | 91.0 | 91.0 | 91.0 | 91.0 | 91.0 |
| **Price (USD/gal)** | $ 2.336 | $. 2.078 | $. 1.934 | $. 1.787 | $. 1.636 | $. 1.518 |

# El Salvador – Premium – Constant Octane



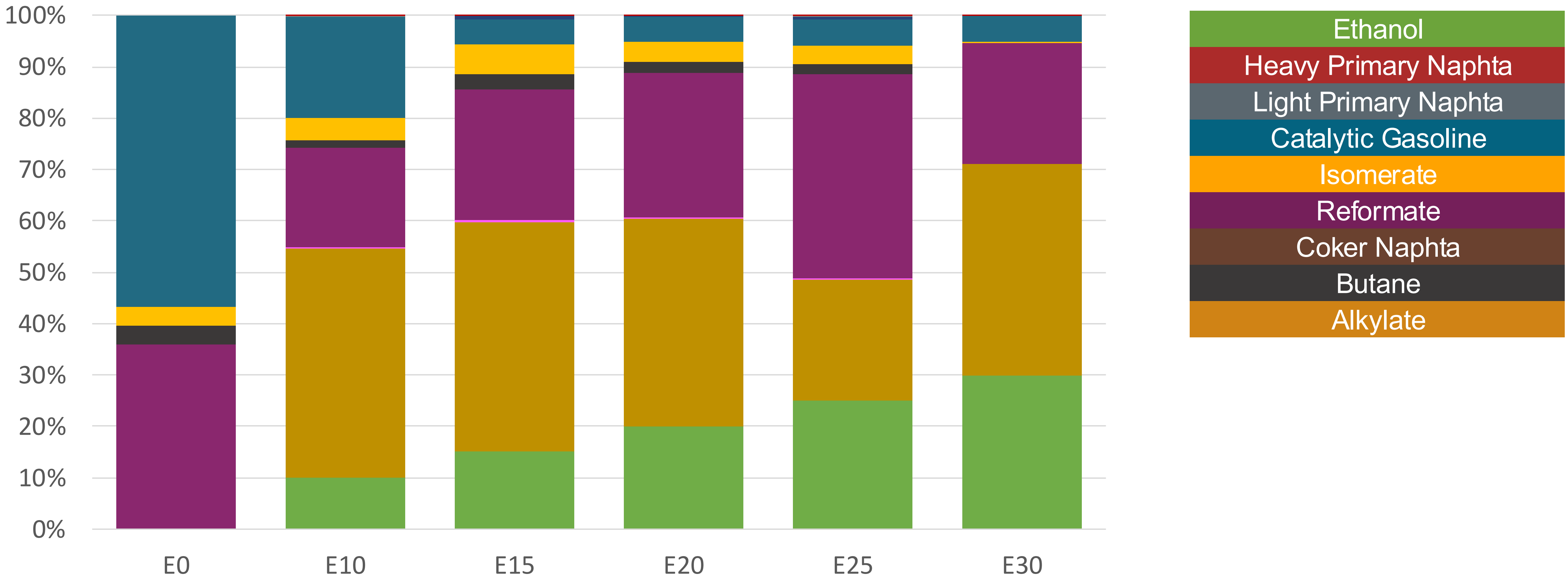

| | | | | | | |
|---|---|---|---|---|---|---|
| **Octane (RON)** | 95.0 | 95.0 | 95.0 | 95.0 | 95.0 | 95.0 |
| **Price (USD/gal)** | $ 2.586 | $ 2.283 | $ 2.149 | $ 2.007 | $ 1.857 | $ 1.709 |

# El Salvador – Regular – Increased Octane




| | | | | | | |
|---|---|---|---|---|---|---|
| **Octane (RON)** | 91.0 | 94.6 | 96.6 | 99.1 | 101.2 | 102.4 |
| **Price (USD/gal)** | $ 2.336 | $ 2.336 | $ 2.336 | $ 2.336 | $ 2.336 | $ 2.336 |

# El Salvador – Premium – Increased Octane




| | E0 | E10 | E15 | E20 | E25 | E30 |
|---|---|---|---|---|---|---|
| **Octane (RON)** | 95.0 | 98.0 | 100.3 | 102.6 | 104.9 | 105.8 |
| **Price (USD/gal)** | $ 2.586 | $ 2.586 | $ 2.586 | $ 2.586 | $ 2.586 | $ 2.586 |

# Vehicle Emission Impact for Ethanol Gasoline Blending

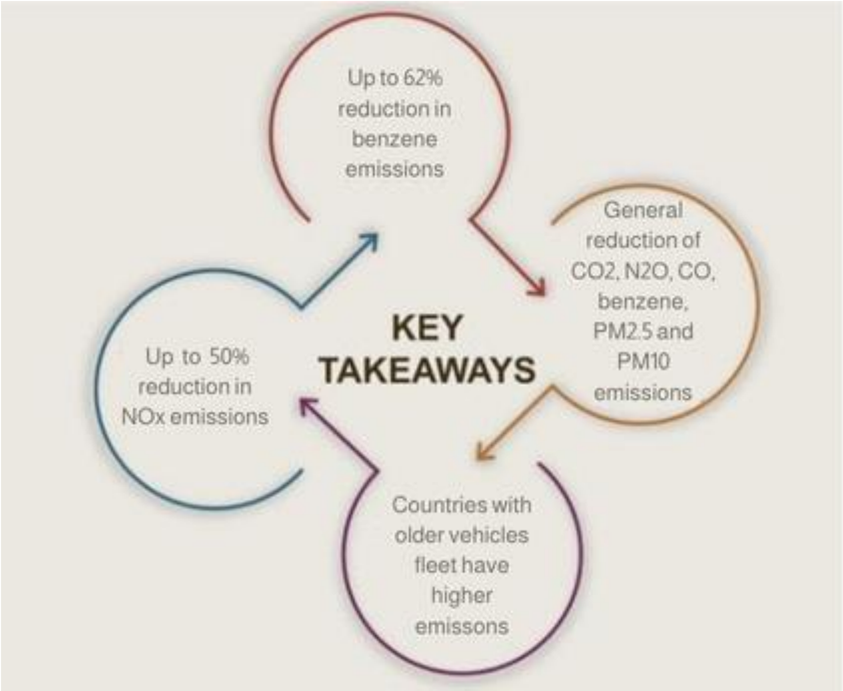
The model used in this analysis takes as a reference the [International Vehicle Emissions Model (IVE).]
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

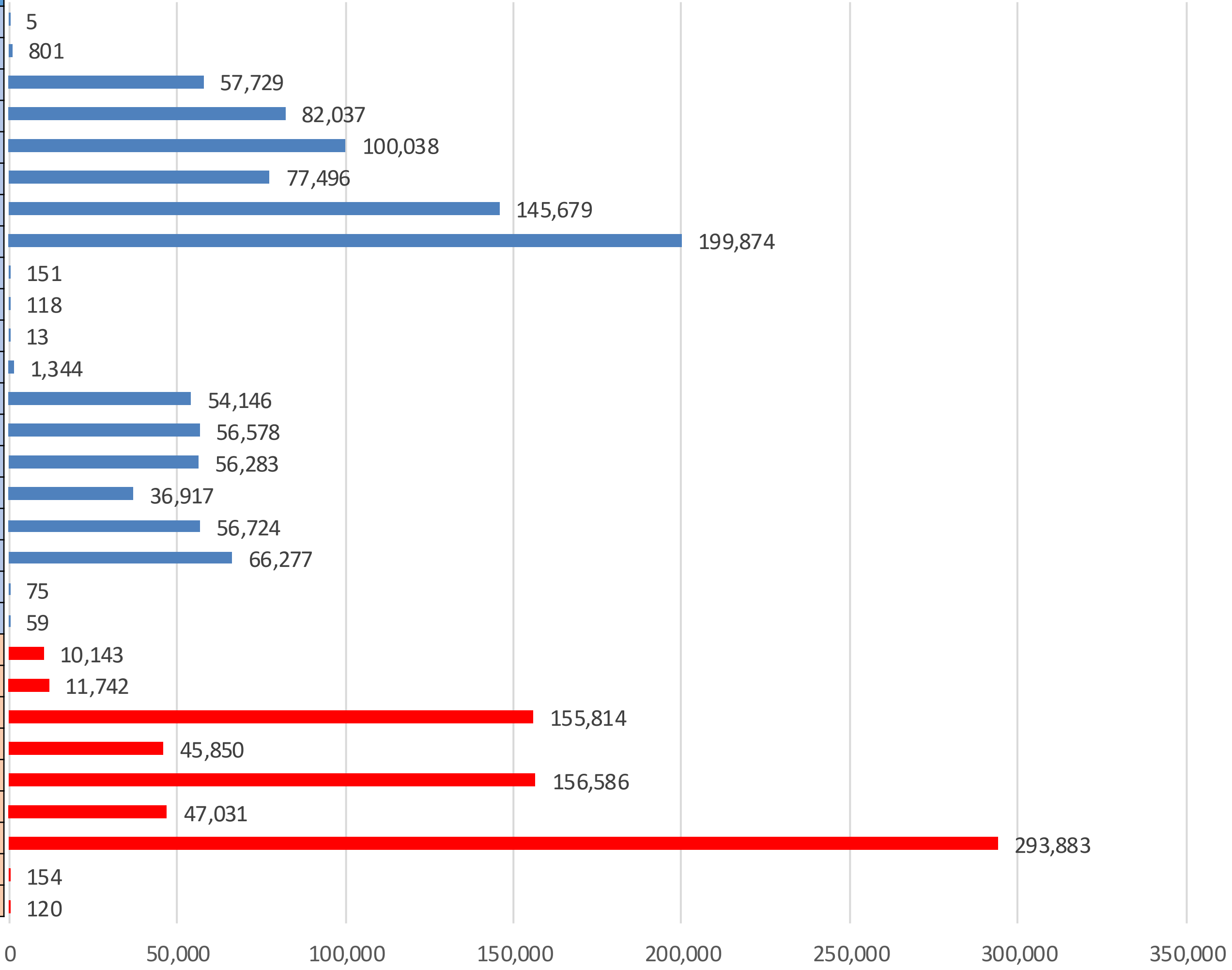
## Main Results




*Source: Bureau of transportation statistics.*

# Gasoline Vehicular Fleet in El Salvador

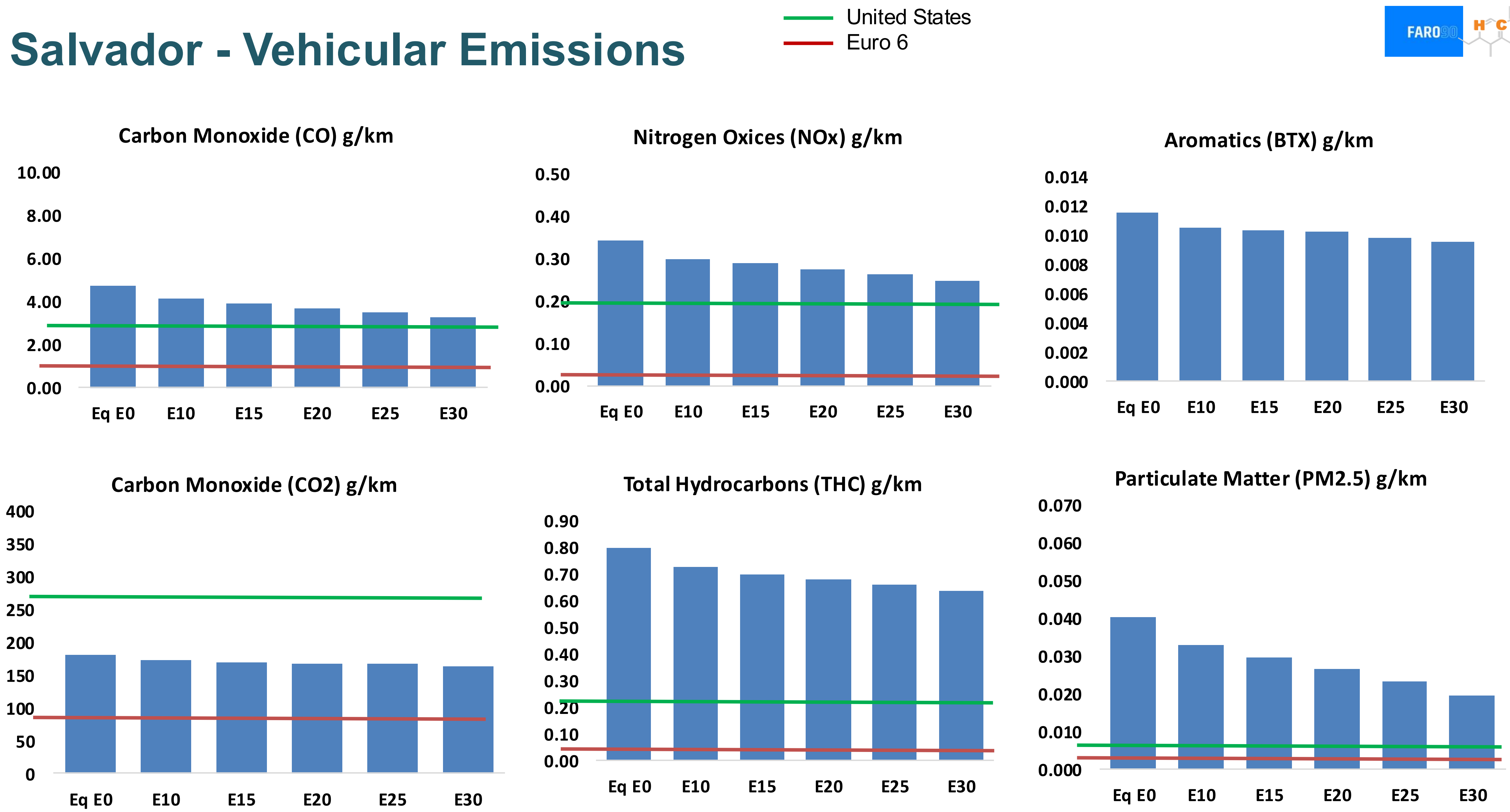
| Type | Motor | Exhaust | Recuperator | Age |
|---|---|---|---|---|
| Auto/Light | Carburator | 3-Ways | PCV Tank | <8 years, < 160 mkm |
| | Single FI-Pt | 3-Ways | | <8 years, < 160 mkm |
| | Multi FI-Pt | 3-Ways | | <8 years, < 160 mkm |
| | | Euro III | | <8 years, < 160 mkm |
| | | Euro IV | | <8 years, < 160 mkm |
| | | Euro V / Euro 6 | | <8 years, < 160 mkm |
| | | | | 4-8 years, 80-160 mkm |
| | | | | >4years, >80 mkm |
| | Hybrid | | | All |
| | Electric | | | All |
| Auto/Medium | Carburator | 3-Ways | PCV Tank | <8 years, < 160 mkm |
| | Single FI-Pt | 3-Ways | | <8 years, < 160 mkm |
| | Multi FI-Pt | 3-Ways | | <8 years, < 160 mkm |
| | | Euro III | | <8 years, < 160 mkm |
| | | Euro IV | | <8 years, < 160 mkm |
| | | Euro V / Euro 6 | | <8 years, < 160 mkm |
| | | | | 4-8 years, 80-160 mkm |
| | | | | >4years, >80 mkm |
| | Hybrid | | | All |
| | Electric | | | All |
| Small Engines | FI 4-cycle | None | PCV | <8 years, < 160 mkm |
| | | Euro II | | <8 years, < 160 mkm |
| | | Euro III | | <8 years, < 160 mkm |
| | | Euro IV | | <8 years, < 160 mkm |
| | | | | 4-8 years, 80-160 mkm |
| | | Euro V | | 4-8 years, 80-160 mkm |
| | | | | >4years, >80 mkm |
| | Hybrid | | | All |
| | Electric | | | All |




Vehicular Fleet: **1,712,991**
Average Age: **9 years**
Motorcycles: **33%**

Source: Registro Publico, Faro 90

# El Salvador - Vehicular Emissions




Source: Faro90

# El Salvador - Vehicular Emissions

| Vehicular Fleet Emissions (kg/day) | Eq E0 | E5 | E10 | E15 | E20 | E25 | E30 | Reduction (%) | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | E10 | E20 |
| CO | 159,777 | 149,748 | 139,718 | 131,939 | 124,313 | 118,510 | 111,443 | -13% | -22% |
| CO2 | 6,127,680 | 5,970,111 | 5,821,296 | 5,704,257 | 5,646,342 | 5,621,675 | 5,518,052 | -5% | -8% |
| VOC | 26,993 | 25,768 | 24,542 | 23,704 | 22,937 | 22,370 | 21,532 | -9% | -15% |
| NOx | 11,643 | 10,750 | 10,130 | 9,811 | 9,300 | 8,897 | 8,425 | -13% | -20% |
| SOx | 70 | 64 | 59 | 55 | 50 | 46 | 41 | -15% | -28% |
| NH3 | 2,338 | 2,315 | 2,292 | 2,303 | 2,329 | 2,347 | 2,362 | -2% | 0% |
| N2O | 74 | 74 | 73 | 74 | 75 | 76 | 77 | -1% | 2% |
| CH4 | 6,335 | 6,335 | 6,335 | 6,430 | 6,569 | 6,689 | 6,803 | 0% | 4% |
| Lead | - | - | - | - | - | - | - | 0% | 0% |
| Butadiene | 132 | 124 | 116 | 110 | 104 | 100 | 93 | -12% | -21% |
| Acetaldehyde | 369 | 423 | 621 | 946 | 1,289 | 1,473 | 1,740 | 68% | 249% |
| Formaldehyde | 1,456 | 1,415 | 1,650 | 1,937 | 2,029 | 2,245 | 2,444 | 13% | 39% |
| Benzene | 392 | 377 | 357 | 352 | 349 | 333 | 323 | -9% | -11% |
| PM2.5 | 376 | 342 | 307 | 277 | 248 | 216 | 183 | -18% | -34% |
| PM10 | 980 | 891 | 802 | 723 | 647 | 564 | 478 | -18% | -34% |



Source: Faro90

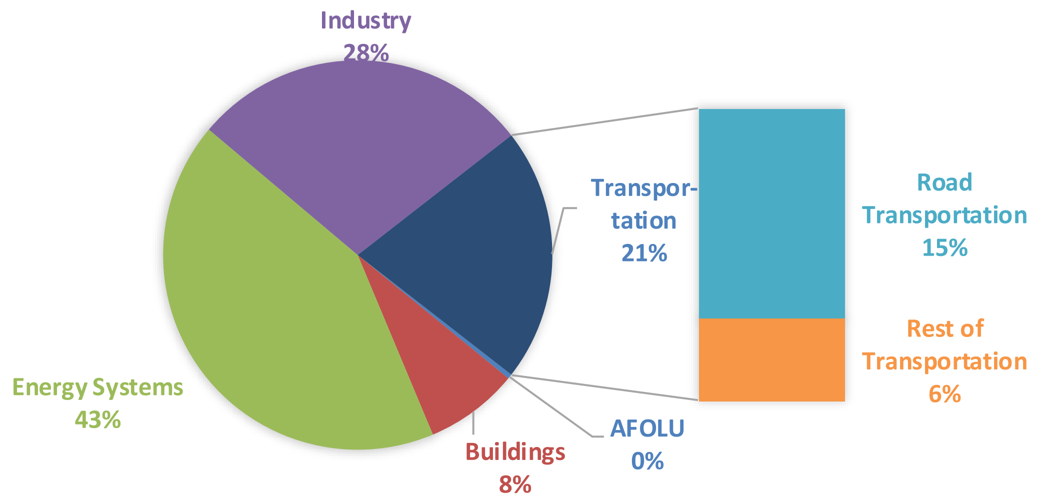
# Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

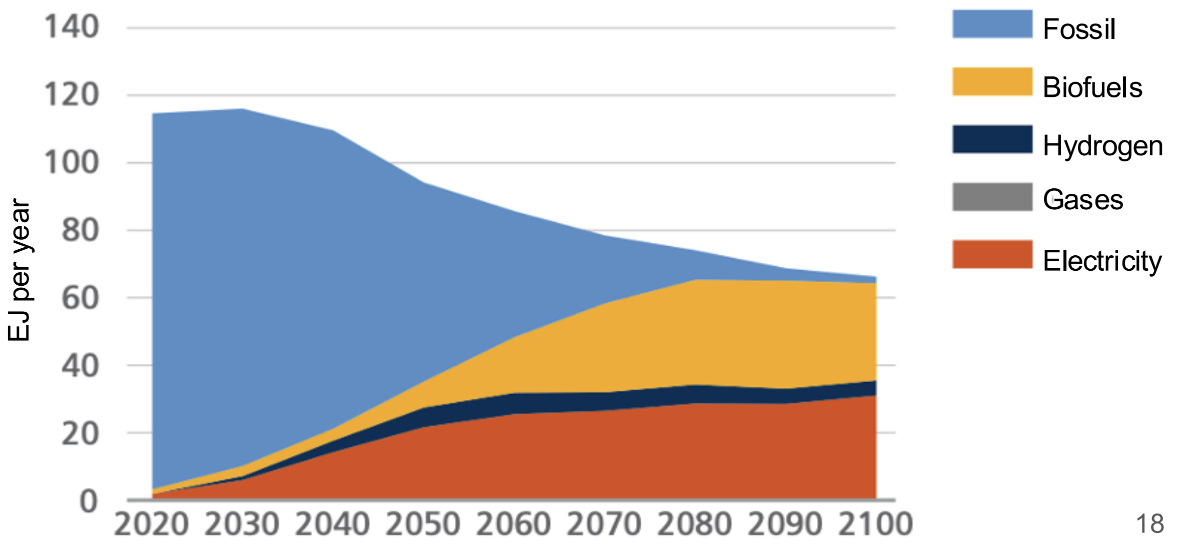
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

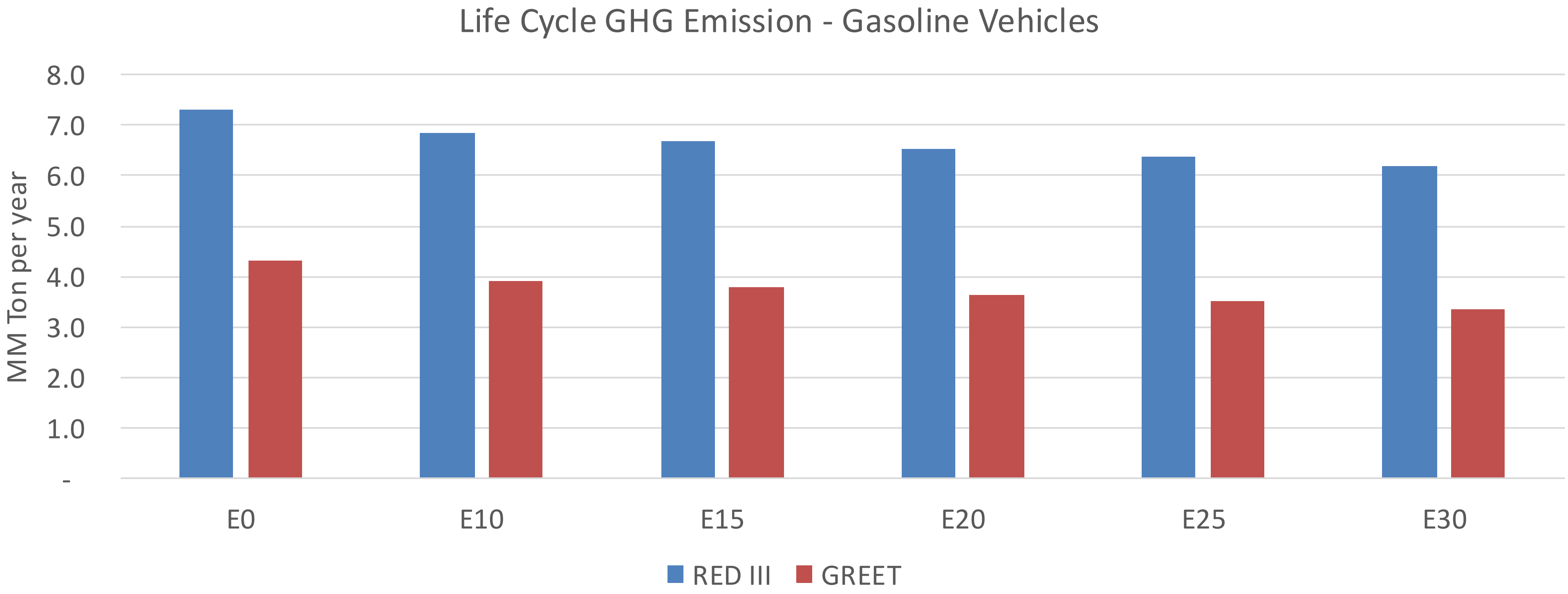
**Current GHG Global Emissions Distribution**




**IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023**



# El Salvador – Gasoline Vehicle Fleet GEI Emissions




| | |
|---|---|
| Country Emissions 2024 (MMT/year) | 14.8 |
| 2035 Target (MMT/year) | 10.3 |
| Delta | 4.4 |

| %Reduction vs E0 | E0 | E10 | E15 | E20 | E25 | E30 |
|---|---|---|---|---|---|---|
| GREET 2024 | 0.0% | 9.3% | 12.6% | 15.9% | 18.7% | 22.4% |
| RED II/III | 0.0% | 6.2% | 8.5% | 10.7% | 12.7% | 15.1% |

| % 2035 Target | E0 | E10 | E15 | E20 | E25 | E30 |
|---|---|---|---|---|---|---|
| GREET 2024 | 0.0% | 9.0% | 12.2% | 15.5% | 18.2% | 21.8% |
| RED II/III | 0.0% | 10.3% | 14.0% | 17.7% | 20.9% | 24.9% |

Source: Greet, RED II/III, climateactiontracker.org, F90